

Name of the Student	U. Somesh Kumar
Reg. No	192425019
GitHub Link	https://github.com/someshkumar12345/Deep-Learning-MLA0403

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Duration: 90 minutes

Assessment Tool Weightage: 30%

Mark : 25

Code of Conduct:

I U. Somesh Kumar (192425019) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

U. Somesh Kumar

U. Somesh Kumar

Signature of the student:

Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

QUESTIONS: 01

Total Marks:25

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

Answer:

1. Article Details

Details	Information
Title	<i>A sentiment analysis framework using RoBERTa with Bi-LSTM classification on Dynamic Twitter data</i>
Authors	Nandini S and Mohan Kabadi
Journal	ETRI Journal
Volume/Issue	Volume 48, Issue 2, pp. 287–303
Year	2025
Publication Date	7 August 2025
DOI	10.4218/etrij.2024-0217
Topic	Natural Language Processing, RoBERTa, BiLSTM, Sentiment Analysis
Data Type	Twitter/social-media text
Main Technique	RoBERTa + Bidirectional LSTM

The selected article proposes a hybrid **Revertability** framework for sentiment analysis of Twitter data. RoBERTa is used to extract semantic and contextual features, while BiLSTM classifies the extracted features into **positive, neutral and negative** sentiment classes. The study was published in 2025, making it a recent research article suitable for this assessment.

Publication link: [Research article – ETRI Journal](#)

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

Answer:

Social media platforms such as Twitter contain a very large amount of user-generated text. People express their opinions, emotions and experiences through short posts. Manually analyzing millions of tweets is difficult and time-consuming. Therefore, an automatic sentiment analysis system is required to classify tweets into different sentiment categories.

The major problem addressed by the article is the difficulty of obtaining **accurate semantic and contextual representations** from Twitter text.

Existing machine-learning and deep-learning approaches may suffer from:

- Difficulty understanding context.
- Difficulty handling short and informal Twitter text.
- Inadequate representation of emotions and opinions.
- Overfitting during training.
- Difficulty capturing relationships between words.
- High computational complexity in some existing approaches.

The authors therefore proposed a combination of **RoBERTa and BiLSTM**. RoBERTa extracts contextual and semantic features, while BiLSTM processes the sequence in both directions.

Why is this problem important?

Sentiment analysis can be used for:

- Customer feedback analysis
- Product reviews
- Social-media monitoring
- Public opinion analysis
- Brand monitoring
- Election/public-policy analysis
- Healthcare and pandemic-related opinion analysis

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

Answer:

The main objectives are:

1. To develop an effective sentiment-analysis framework for Twitter data.
2. To extract meaningful semantic and contextual features from tweets.
3. To use **RoBERTa** for feature extraction.
4. To use **BiLSTM** for sentiment classification.
5. To classify tweets into positive, neutral and negative categories.
6. To improve model convergence using **Adam optimization and Reduced Learning Rate on Plateau (RLRP)**.
7. To reduce overfitting during training.
8. To compare the proposed model with existing sentiment-analysis approaches.
9. To investigate whether the model can be used for real-time Twitter sentiment analysis.

The authors specifically designed the framework to combine the strong contextual representation of RoBERTa with the bidirectional sequence-processing ability of BiLSTM.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

Answer:

The research uses three datasets.

4.1 Sentiment140 Dataset

The Sentiment140 dataset contains approximately **1.6 million annotated tweets**.

The dataset contains positive and negative tweets and was collected using the Twitter Search API. The paper reports approximately **800,000 positive and 800,000 negative tweets**.

4.2 COVID-19 Dataset

The COVID-19 dataset contains **3,090 tweets** collected from Indian Twitter users during the COVID-19 curfew.

The tweets are related to topics such as:

- COVID-19
- Coronavirus
- Lockdown
- Public reactions to the pandemic

4.3 Twitter Emoji Dataset

The third dataset contains Twitter text and emoji information. It was used to study how emojis can represent emotions in social-media communication.

Data type

The data is primarily:

Text + social-media/emoji information

Dataset processing

The tweets undergo preprocessing before being given to the model.

The paper describes:

- Tokenization
- Stop-word removal
- Stemming
- Punctuation removal
- Acronym expansion

These operations reduce noise and make the text easier for the model to process.

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.
Describe the data preprocessing techniques used.
Explain the training procedure and important parameters, wherever available.

Answer:

The overall methodology consists of the following stages:

Step 1 – Input

Raw Twitter posts are collected.

Step 2 – Preprocessing

The raw tweets are cleaned using:

- Tokenization
- Stop-word removal
- Stemming
- Punctuation removal
- Acronym expansion

Step 3 – Feature Extraction

The cleaned text is given to **RoBERTa**.

RoBERTa uses transformer-based self-attention to understand relationships between words and generate contextual representations.

Step 4 – BiLSTM Classification

The features extracted by RoBERTa are passed to a **Bidirectional LSTM**.

BiLSTM processes the sequence:

- Forward direction
- Backward direction

Therefore, it can use information from both previous and subsequent words.

Step 5 – Dense Layer

The BiLSTM output is given to a dense classification layer.

Step 6 – Softmax

Softmax produces probabilities for:

- Negative
- Neutral
- Positive

Step 7 – Optimization

The authors use:

- Adam optimizer
- Reduced Learning Rate on Plateau
- Dropout
- L2 regularization
- Early stopping

The reported tuned parameters include dropout = **0.5**, batch size = **32**, initial learning rate = **0.001**, and **10 epochs**.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Answer:

What is LSTM?

LSTM stands for **Long Short-Term Memory**.

It is a special type of Recurrent Neural Network designed to process sequential data.

Traditional RNNs can suffer from the vanishing-gradient problem when processing long sequences.

LSTM addresses this using gates:

- Forget gate
- Input gate
- Output gate

These gates control what information should be:

- Forgotten
- Stored
- Passed to the next step

What is BiLSTM?

BiLSTM = Bidirectional Long Short-Term Memory

Instead of processing the input only from left to right, BiLSTM processes it in two directions:

Forward:

Word 1 → Word 2 → Word 3 → Word 4

Backward:

Word 4 → Word 3 → Word 2 → Word 1

The outputs from both directions are combined.

Advantage

This allows the model to understand the context from both the **previous and following words**.

For example:

"The movie was not good"

The meaning of "good" depends strongly on the word "not". Bidirectional processing can provide better contextual information.

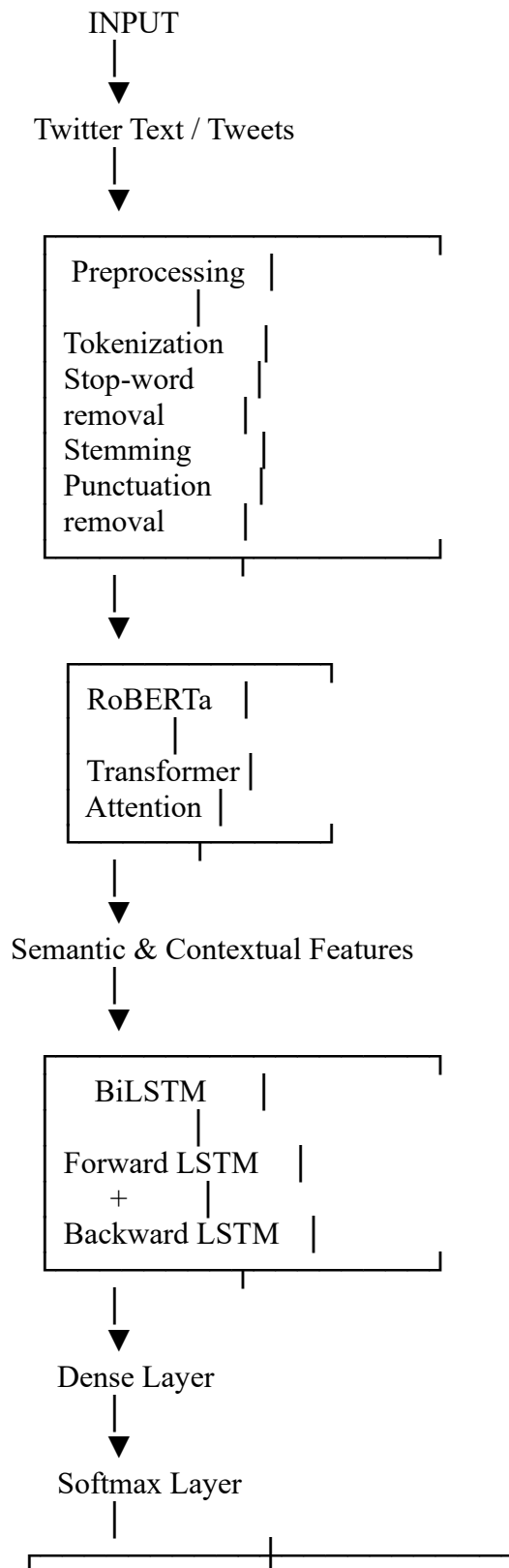
The selected paper uses BiLSTM after RoBERTa feature extraction to classify sentiment.

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:
Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →
Classification/Decoder → Output
Identify the role of the major components in the architecture.

Answer:

The simplified architecture of the proposed model is:



▼ ▼ ▼
Negative Neutral Positive

Role of each component

Preprocessing: Removes unnecessary information from tweets.

RoBERTa: Extracts semantic and contextual features using transformer-based attention.

BiLSTM: Processes the extracted sequence in both forward and backward directions.

Dense layer: Converts the learned representation into classification information.

Softmax: Calculates the probability of each sentiment class.

The paper describes the same general pipeline: preprocessing → RoBERTa feature extraction → BiLSTM classification → softmax output.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score, Loss, Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

Answer:

The authors use the following evaluation metrics:

1. Accuracy

Measures the percentage of correctly classified samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision

Measures how many predicted positive samples are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

3. Recall

Measures how many actual positive samples are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

4. F1-Score

Combines precision and recall.

$$F1 = \frac{2(Precision)(Recall)}{Precision + Recall}$$

These four metrics were used by the authors to evaluate the model.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

Answer:

The proposed RoBERTa + BiLSTM model achieved strong results.

Sentiment140 Dataset

Metric	Proposed Model
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Accuracy	97.905%
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Precision	97.921%
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Recall	97.994%
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F1-Score	97.983%
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COVID-19 Dataset

The proposed model achieved an accuracy of approximately **96.93%**.

Comparison on Sentiment140

Model	Accuracy
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GARN	97.864%
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RoBERTa-GRU	89.590%
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ARO-RRVFLN	96.050%
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ASASM-HHODL	84.250%
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RoBERTa + BiLSTM	97.905%
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The proposed model produced the highest accuracy among the listed comparison methods.

Discussion

The strong performance can be attributed to the combination of:

1. RoBERTa's contextual feature extraction.
2. BiLSTM's bidirectional sequence processing.
3. Adam optimization.
4. Learning-rate scheduling.
5. Dropout and regularization.
6. Early stopping.

The combination allows the system to understand both the semantic meaning and sequence context of tweets.

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

Answer:

1. Better contextual understanding

RoBERTa can understand relationships between words using transformer-based self-attention.

2. Bidirectional processing

BiLSTM considers information from both directions, which improves understanding of sentence context.

3. High classification performance

The model achieved approximately **97.9% accuracy** on the Sentiment140 dataset.

4. Reduced overfitting

Dropout, L2 regularization, early stopping and learning-rate reduction help control overfitting.

5. Real-time applicability

The paper describes deployment using Docker, Flask REST APIs and Microsoft Azure, showing potential for practical deployment.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

Answer:

Although the proposed model achieves high performance, some limitations can be identified.

1. High computational requirement

RoBERTa combined with BiLSTM requires significant computational resources.

2. Model complexity

The architecture is more complicated than a simple RNN or LSTM.

3. Training time

Training a transformer-based model followed by BiLSTM can require considerable time.

4. Dependency on Twitter data

The model is mainly evaluated on social-media text, so performance may differ on other domains.

5. Informal language

Twitter contains slang, abbreviations, emojis and spelling errors, which can still create difficulties.

6. Generalization

Although multiple datasets are used, additional testing across languages, countries and domains would strengthen the conclusions.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

Answer:

Is BiLSTM appropriate for this problem?

Yes.

Twitter sentiment analysis is a sequential NLP problem. The meaning of a word can depend on words appearing before and after it. Therefore, BiLSTM is suitable because it processes information in both directions.

Is the dataset adequate?

The Sentiment140 dataset is large, with approximately **1.6 million tweets**, which is a strong advantage. However, the COVID-19 dataset contains only 3,090 tweets, so additional datasets would improve generalization.

Are the evaluation metrics appropriate?

Yes. Accuracy, precision, recall and F1-score are appropriate for sentiment classification. F1-score is particularly useful because it considers both precision and recall.

Are the experimental results convincing?

The results are convincing because the proposed model is compared against several existing methods and performs strongly. However, more extensive cross-domain and multilingual testing would make the evaluation stronger.

Could another technique provide better performance?

Yes.

A pure transformer such as **BERT/RoBERTa fine-tuning** could potentially provide competitive or better performance while avoiding the additional recurrent layer. However, the combination of RoBERTa + BiLSTM provides a strong hybrid architecture because RoBERTa extracts contextual representations while BiLSTM explicitly models bidirectional sequential information.

Overall critical opinion

The proposed architecture is technically appropriate and achieves excellent results. The major trade-off is **accuracy versus computational complexity**.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Answer:

The following improvements can be considered:

1. Develop a **multilingual sentiment-analysis model**.
2. Include more recent Twitter/X data.
3. Test the model on other social-media platforms.
4. Apply transfer learning for domain-specific sentiment analysis.
5. Reduce model size for mobile and edge deployment.
6. Use model quantization to improve inference speed.
7. Investigate lightweight transformer models.
8. Improve emoji and slang understanding.
9. Develop explainable AI techniques to show why a tweet is classified as positive or negative.
10. Continuously retrain the model using new social-media data.

The authors also suggest future work involving domain adaptation, transfer learning and feature selection to improve applicability and reduce feature dimensionality

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

Answer:

The reviewed research presents a **RoBERTa-BiLSTM framework for Twitter sentiment analysis**.

The system first preprocesses Twitter text and uses **RoBERTa** to extract semantic and contextual features. These features are then passed to **BiLSTM**, which processes the information in both forward and backward directions. Finally, a dense and softmax layer classifies the tweet into positive, neutral or negative sentiment.

The model achieved **97.905% accuracy, 97.921% precision, 97.994% recall and 97.983% F1-score on the Sentiment140 dataset**. It also achieved **96.93% accuracy on the COVID-19 dataset**.

Overall, the article provides a strong example of applying advanced deep-learning techniques to a real-world NLP problem. The main strength is its combination of **transformer-based contextual feature extraction and bidirectional recurrent processing**. The main limitation is the relatively high computational complexity.

Therefore, **RoBERTa + BiLSTM is an effective approach for Twitter sentiment classification**, although future work should focus on lightweight models, multilingual datasets, explainability and broader domain testing.

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

Answer:

Main Research Article

Nandini, S., & Kabadi, M. (2025). **A sentiment analysis framework using RoBERTa with Bi-LSTM classification on Dynamic Twitter data**. *ETRI Journal*, 48(2), 287–303. DOI: 10.4218/etrij.2024-0217.

Additional References

1. Liu, Y., et al. (2019). **RoBERTa: A Robustly Optimized BERT Pretraining Approach**. arXiv preprint arXiv:1907.11692.
2. Hochreiter, S., & Schmidhuber, J. (1997). **Long Short-Term Memory**. *Neural Computation*, 9(8), 1735–1780.
3. Schuster, M., & Paliwal, K. K. (1997). **Bidirectional recurrent neural networks**. *IEEE Transactions on Signal Processing*, 45(11), 2673–2681.

4. Go, A., Bhayani, R., & Huang, L. (2009). **Twitter Sentiment Classification using Distant Supervision**. Stanford University.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis